



Master in Computer Vision | Barcelona

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Barcelona**



Module: 3D Vision

Project: 3D recovery of urban scenes (Session 4)

Original Lab: Marc Perez (marc.perez.quintana@upc.edu)

Goal

3D Reconstruction from uncalibrated images with a stratified method, by applying a Structure from Motion (SfM) pipeline in order to achieve a 3D sparse reconstruction. Dense 3D Reconstruction with Multi-View Stereo.



**Sparse Reconstruction
(Point cloud)**

Images to 3d
→



Dense reconstruction (mesh)

Colmap

<https://github.com/colmap/colmap>

- Opensource (BSD license)
- Pre-built binaries for Ubuntu/Windows/Mac
- Widely used

2023

Watch 172 Fork 1.2k Star 4.7k

Last year

Watch 175 Fork 1.4k Star 6.2k

This year

Watch 174 Fork 1.6k Star 8k

```
@inproceedings{schoenberger2016sfm,
  author={Sch\"{o}nberger, Johannes Lutz and Frahm, Jan-Michael},
  title={Structure-from-Motion Revisited},
  booktitle={Conference on Computer Vision and Pattern Recognition (CVPR)},
  year={2016},
}

@inproceedings{schoenberger2016mvs,
  author={Sch\"{o}nberger, Johannes Lutz and Zheng, Enliang and Pollefeys, Marc and Frahm, Marc},
  title={Pixelwise View Selection for Unstructured Multi-View Stereo},
  booktitle={European Conference on Computer Vision (ECCV)},
  year={2016},
}
```

- Documented

<https://colmap.github.io/>

<https://demuc.de/tutorials/cvpr2017/>

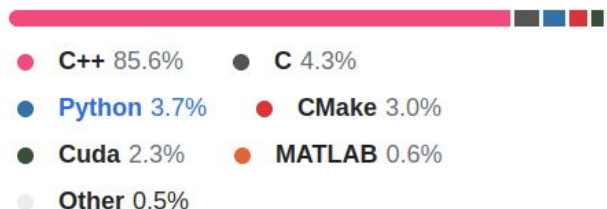
- Maintained

last week 2,013 Commits

- Code in C/C++

Cited by 6491
(last year: 4396,
2023: 2892)

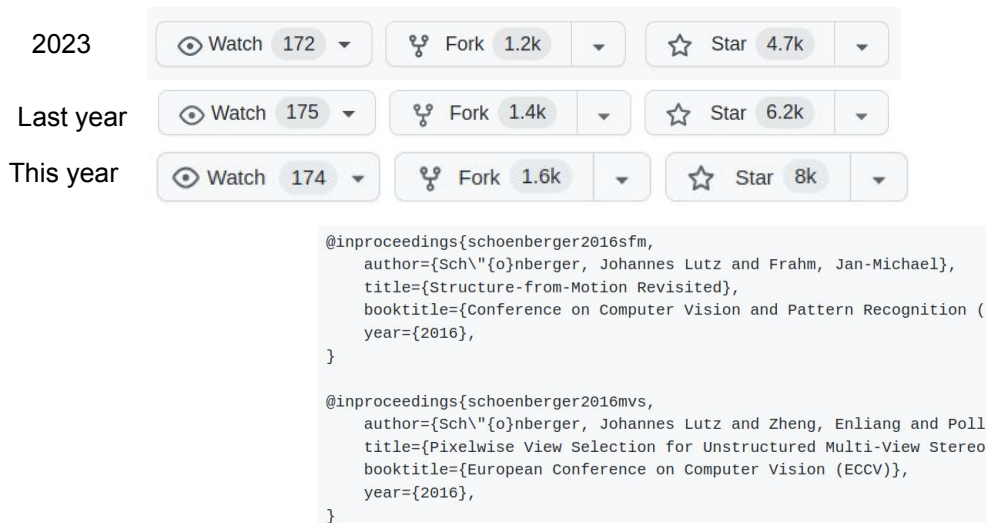
Cited by 2548
(last year: 1777,
2023: 1139)



Colmap

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<https://colmap.github.io/>
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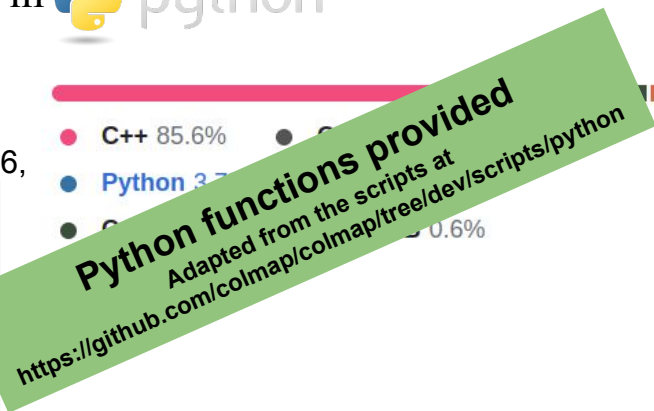
- Maintained



- Code in  python™

Cited by 6491
(last year: 4396,
2023: 2892)

Cited by 2548
(last year: 1139,
2023: 1139)



Colmap

- Feedback from 2023
 - “Don't ever use colmap again **there's better alternatives**”

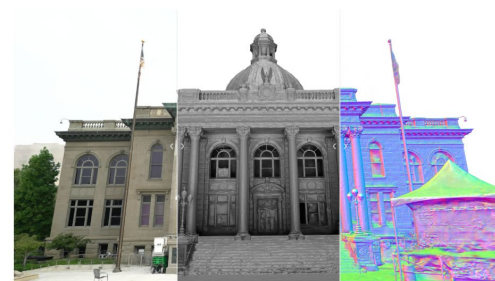
Neuralangelo: High-Fidelity Neural Surface Reconstruction

Zhaoshuo Li^{1,2} Thomas Müller¹ Alex Evans¹ Russell H. Taylor²
Mathias Unberath² Ming-Yu Liu¹ Chen-Hsuan Lin¹

¹NVIDIA Corporation ²Johns Hopkins University

IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2023

TIME's Best Inventions of 2023



Code: <https://github.com/NVlabs/neuralangelo>

Project: <https://research.nvidia.com/labs/dir/neuralangelo/>

Paper: <https://research.nvidia.com/labs/dir/neuralangelo/paper.pdf>

Neuralangelo: High-Fidelity Neural Surface Reconstruction

Zhaoshuo Li^{1,2}Thomas Müller¹Alex Evans¹Russell H. Taylor²Mathias Unberath²Ming-Yu Liu¹Chen-Hsuan Lin¹¹NVIDIA Corporation²Johns Hopkins University

IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2023

TIME's Best Inventions of 2023

neuralangelo / third_party /

Add file

...



Max Li move colmap to tag 3.8

3464428 · 4 months ago

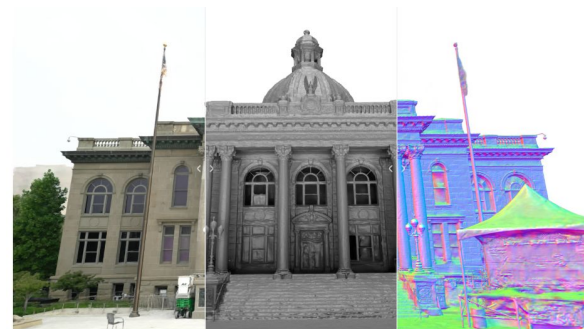
History

Name	Last commit message	Last commit date
..		
colmap @ 43de802	move colmap to tag 3.8	4 months ago

2. Q: The reconstruction of my custom dataset is bad. What can I do?

A: It is worth looking into the following:

- The camera poses recovered by COLMAP may be off. We have implemented tools (using Blender or Jupyter notebook) to inspect the COLMAP results.
- The computed bounding regions may be off and/or too small/large. Please refer to data preprocessing on how to adjust the bounding regions manually.
- The video capture sequence may contain significant motion blur or out-of-focus frames. Higher shutter speed (reducing motion blur) and smaller aperture (increasing focus range) are very helpful.



Files



main



Go to file

t

SIBR_viewers

- > arguments
- > assets
- > gaussian_renderer
- > lpipsPyTorch
- ✓ scene
 - __init__.py
 - cameras.py
 - colmap_loader.py
 - dataset_readers.py
 - gaussian_model.py
- > submodules
- > utils
 - .gitignore
 - .gitmodules
 - LICENSE.md

gaussian-splatting / scene colmap_loader.py



bkerbl Bump rasterizer, faster text loading

ea68bdf · 2 years ago

History

Code

Blame

294 lines (269 loc) · 11.6 KB

Raw



```
1 #
2 # Copyright (C) 2023, Inria
3 # GRAPHDECO research group, https://team.inria.fr/graphdeco
4 # All rights reserved.
5 #
6 # This software is free for non-commercial, research and evaluation use
7 # under the terms of the LICENSE.md file.
8 #
9 # For inquiries contact george.drettakis@inria.fr
10 #
11
12 import numpy as np
13 import collections
14 import struct
15
16 CameraModel = collections.namedtuple(
17     "CameraModel", ["model_id", "model_name", "num_params"])
18 Camera = collections.namedtuple(
19     "Camera", ["id", "model", "width", "height", "params"])
20 BaseImage = collections.namedtuple(
21     "Image", ["id", "qvec", "tvec", "camera_id", "name", "xys", "point3D_ids"])
22 Point3D = collections.namedtuple(
23     "Point3D", ["id", "xyz", "rgb", "error", "image_ids", "point2D_idxs"])
24
```

Symbols

Find definitions and references for functions and other symbols in this file by clicking a symbol below or in the code.

Filter symbols

- const CameraModel
- const Camera
- const BaseModel
- const Point3D
- const CAMERA_MODELS
- const CAMERA_MODEL_IDS
- const CAMERA_MODEL_NAMES
- func qvec2rotmat
- func rotmat2qvec
- class Image
 - func qvec2rotmat
 - func read_next_bytes

Colmap

- Feedback from last year
 - “3d reconstruction with colmap seems **too much research oriented**. A pipeline for the industry, with some techniques that could be run in scripting for dense 3d could be awesome.”
 - “We had to spend some time **installing the software**, It'd [be] good if the students can download a virtual machine with all they need on it.”

Mandatory Tasks

- 1. 3D mesh reconstruction from a set of images from the Gerrard Hall dataset. **(1.5)**
- 2. Step-by-step 3D mesh reconstruction on the CASTLE dataset (lab 4) following the notebook provided, explain each step. **(6.0)**
- 3. Configure the reconstruction to improve the results. **(1.0)**
- 4. Reconstruct a 3D mesh from images captured by you. **(1.5)**

Optional Tasks

- 5. Use Neuralangelo or Gaussian Splatting to reconstruct a 3D mesh from images captured by you **(+2.0)**
 - <https://research.nvidia.com/labs/dir/neuralangelo/>
 - <https://github.com/graphdeco-inria/gaussian-splatting/tree/main>

Assignment

Session 4

- Code is provided in python in a jupyter notebook.
- Deliver before 9 AM of Monday, January 27. (You have some extra time this week)

Deliverables

- **Jupyter notebook:** ready to run.
 - Document your code and decisions on markdown. All code in the notebook! **Not aux/utils files!**
 - Be clear of what information is assumed/required for each algorithm/operation.
 - Understand the equations do not just reproduce them from the slides.
 - **Report:**
 - Short report with in depth analysis, insights, problems and comments.
 - Do not paste code in report. I am interested in analysis and justification.
 - You can use the notebook as a report **IF, AND ONLY IF**, you format the notebook appropriately.
-  **UPLOAD THE NOTEBOOK EXECUTED** and just in case upload it as well as PDF or HTML.



Meshes:

- The **3D meshes** generated in all exercises or videos showing the 3D structure of the mesh. (link to Google Drive for example, if they take too much space)

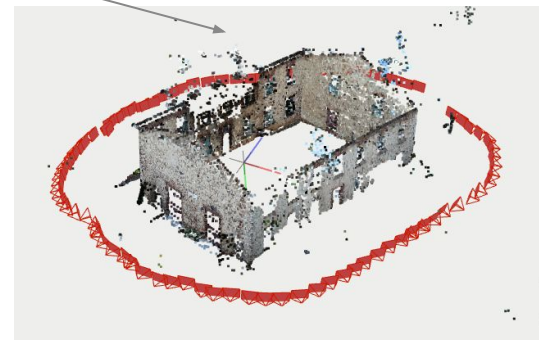
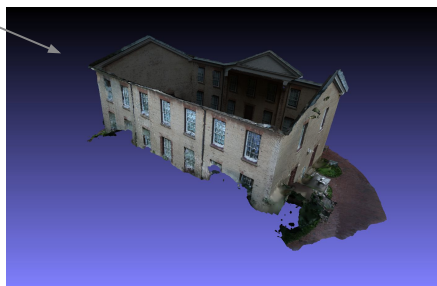
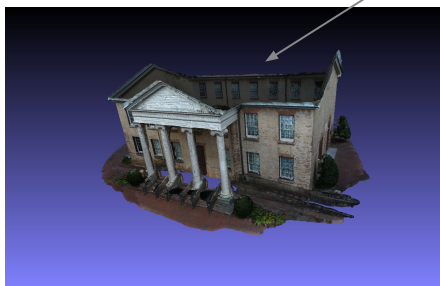


Naming conventions:

- All the files (zip, notebook, report) should be named [**<GROUP_ID>**]**_lab4** (e.g. [1]**_lab4.ipynb**)

1. 3D mesh reconstruction from a set of images from the Gerrard Hall dataset (1.5)

- Install Colmap: <https://colmap.github.io/install.html>
- Read the documentation: <https://colmap.github.io/>, <https://demuc.de/tutorials/cvpr2017/>
- Download the Gerrard Hall dataset: <https://demuc.de/colmap/datasets/>
- Run the automatic reconstruction on the Gerrard Hall dataset
- Visualize the sparse reconstruction on Colmap
(for example on Meshlab: <https://www.meshlab.net/>)

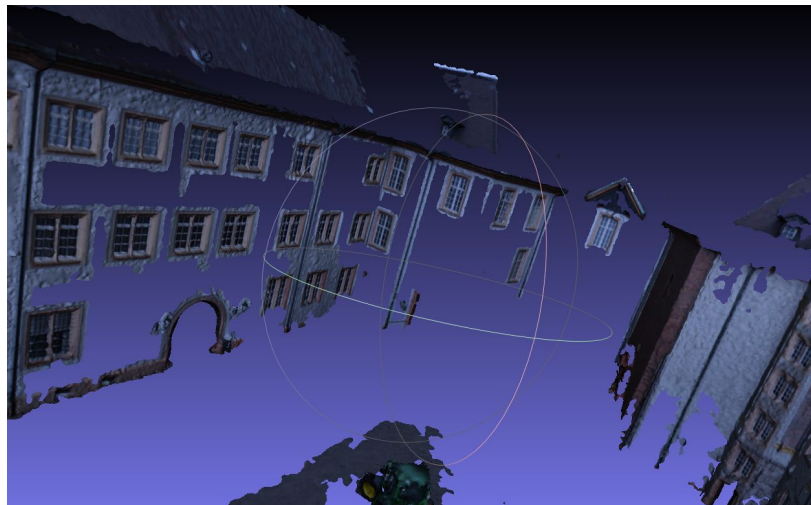


2. Analyze reconstructions using python. (6.0)

- 2.1 Run the notebook, using the Gerrard Hall reconstruction (0.5)
- 2.2 Plot the 3D points coloured according to the number of images and error. (0.5)
- 2.3 Plot the 3D points that originated from a keypoint in the first image. Also plot the image with the keypoints (1.0)
- 2.4 Create a visualization for the number of matches between all images. (1.0)
 - For example: <https://seaborn.pydata.org/generated/seaborn.heatmap.html>
- 2.5 Visualize the keypoints and matches between the two images used in lab 4 using Colmap, how it compares to the results from lab 4? (1.0)
- 2.6 Triangulate and visualize the 3D points from the keypoints extracted using Colmap on the two images used in lab 4, how it compares to the results from lab 4? (1.0)
- 2.7 Visualize the sparse reconstruction using the 2 images from lab 4, and the complete CASTLE dataset. Comment on the differences between techniques and number of images used. (1.0)
 - <https://documents.epfl.ch/groups/c/cv/cvlab-unit/www/data/multiview/denseMVS.html> (Castle P-30)

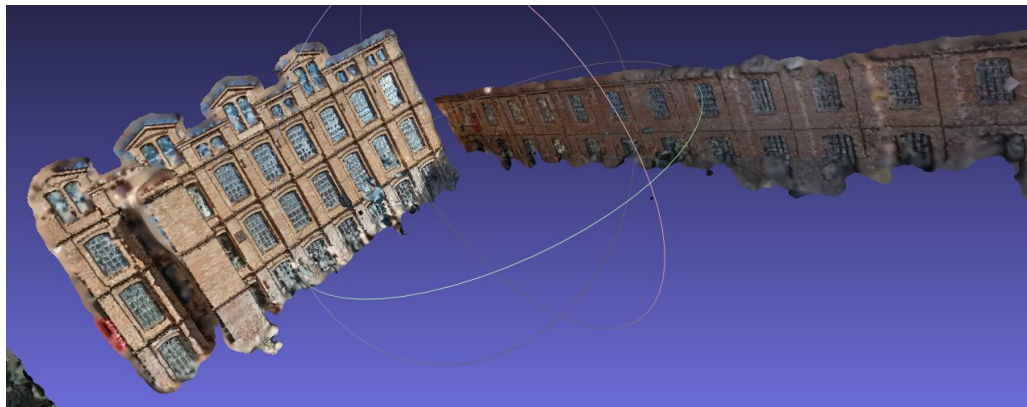
3. Configure the reconstruction to improve the results. (1.0)

- Configure the reconstruction to get the best possible mesh, explain the parameters changed and why you decided to change them. Remove noisy points with Meshlab or others.



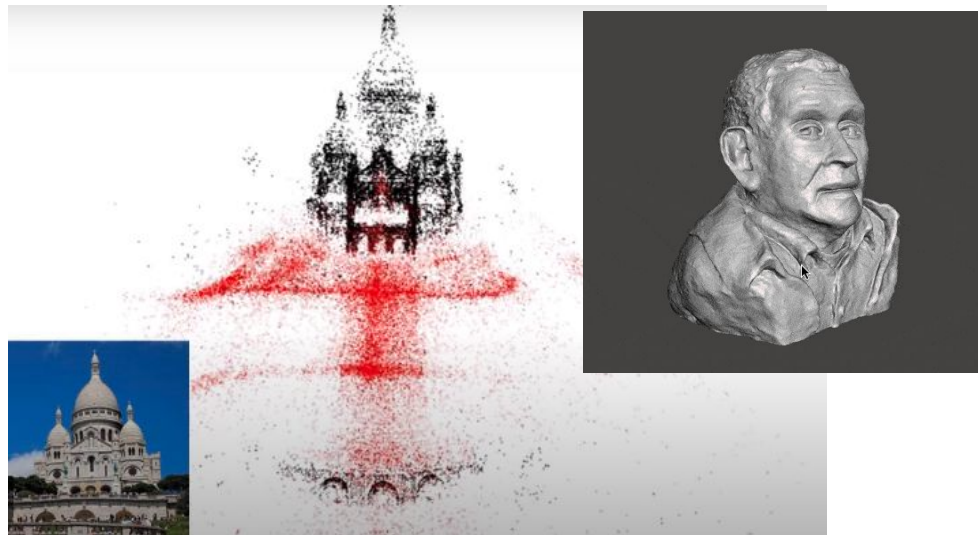
4. Reconstruct a 3D mesh from images captured by you. (1.5)

- Ideally from the same environment, to be able to compare between groups: [UPF campus!](#)
- Image capturing is important to get good results.



4. Reconstruct a 3D mesh from images captured by you. (1.5)

- What are the requirements for image capturing to get good 3D reconstructions?
- You might need to change the configuration to get a good reconstruction, comment on the changes you make.
- You can apply it to other scenes or objects.
- You can try with different cameras.
- You can try with [images from the web](#).



5. Use **Neuralangelo** or Gaussian splatting to reconstruct a 3D mesh from images captured by you (Optional 2.0)

<https://research.nvidia.com/labs/dir/neuralangelo/>

<https://github.com/graphdeco-inria/gaussian-splatting/tree/main>



Image from the paper:

<https://research.nvidia.com/labs/dir/neuralangelo/paper.pdf>

5. Use Neuralangelo or **Gaussian splatting** to reconstruct a 3D mesh from images captured by you (Optional 2.0)

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<https://github.com/graphdeco-inria/gaussian-splatting/tree/main>



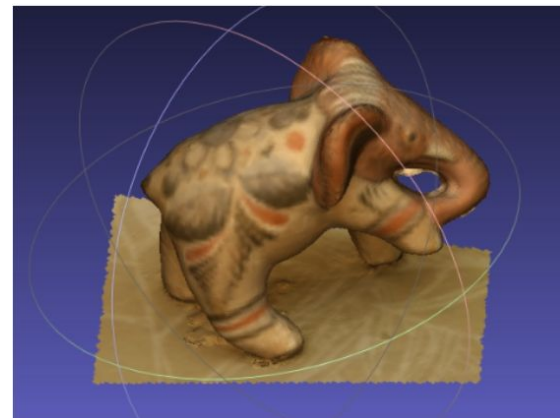
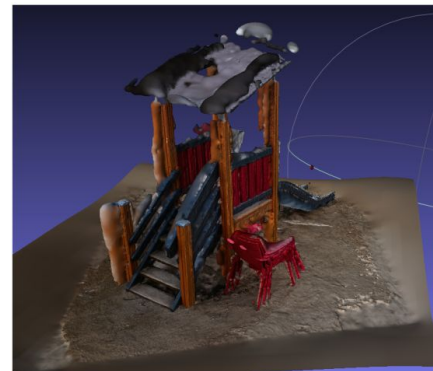
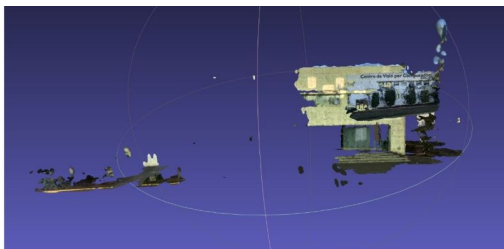
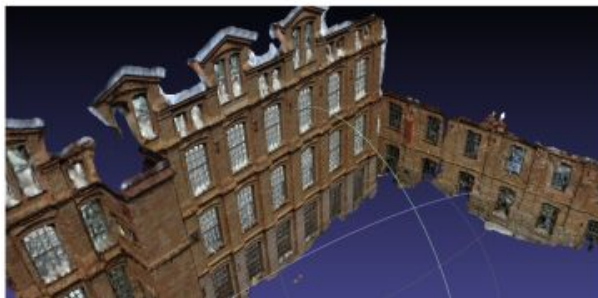
Image from the paper:

https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/3d_gaussian_splattting_high.pdf

Every group can access a GPU?

There is an option to use a cluster, but it has not been tested.
If any group wants to try let me know.

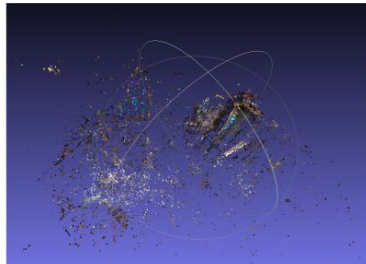
Some 3D reconstructions from 2023



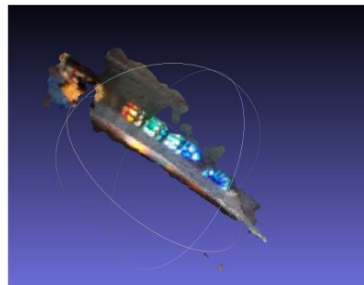
Some 3D reconstructions from 2023



(a) Sample image for the generated dataset of Sagrada Familia from Google StreetView



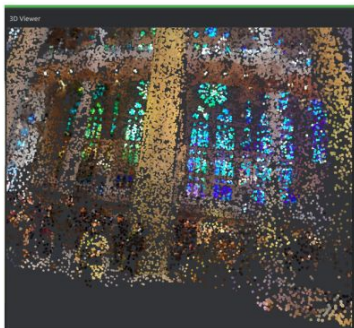
(b) Sparse representation of the data of Sagrada Familia



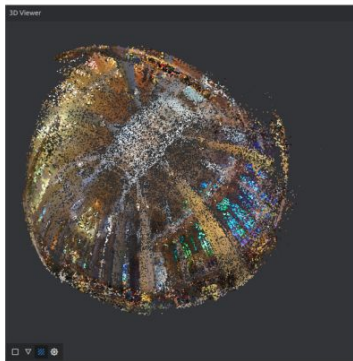
(c) Dense representation for the data of Sagrada Familia



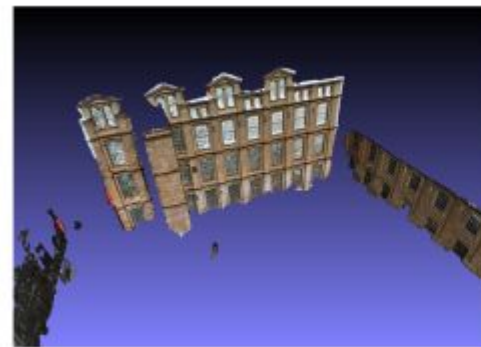
(a) Example of working with Meshroom



(b) Inner view of Sagrada Familia reconstruction by Meshroom.



(c) Bottom view of Sagrada Familia reconstruction.



Some 3D reconstructions from 2024



Neuralangelo



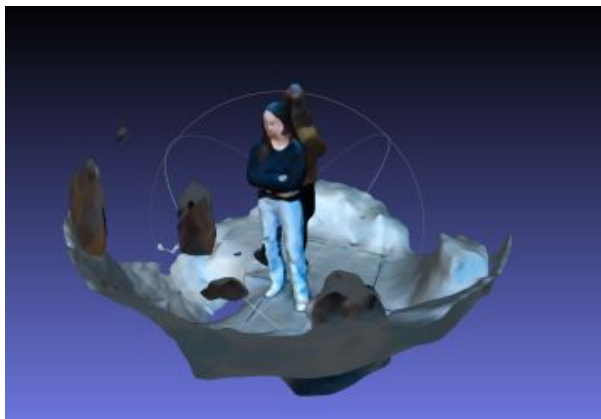
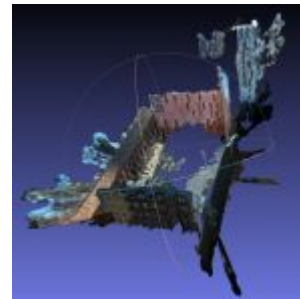
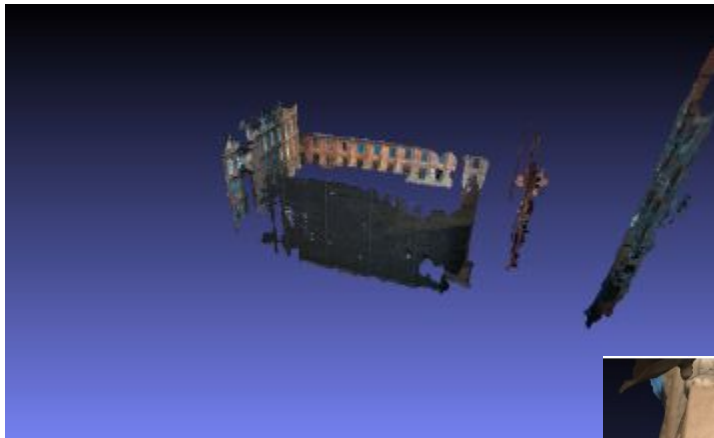
Fig. 22: Neuralangelo reconstruction from 100 epochs



Fig. 23: Neuralangelo reconstruction from 44k epochs

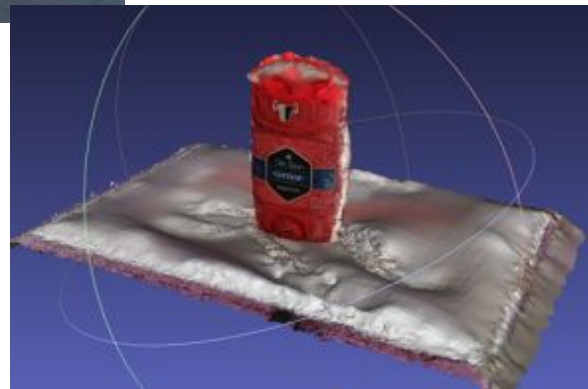
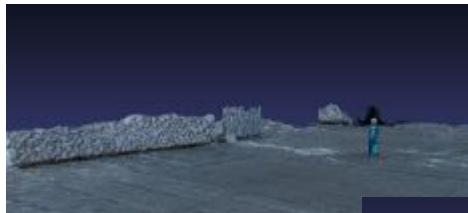
(Provided dataset)

Some 3D reconstructions from 2024

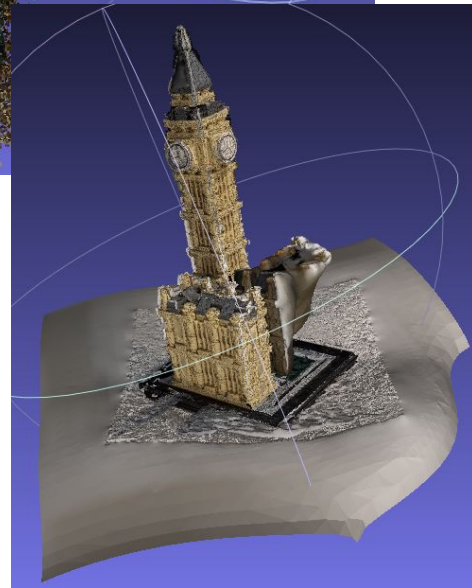
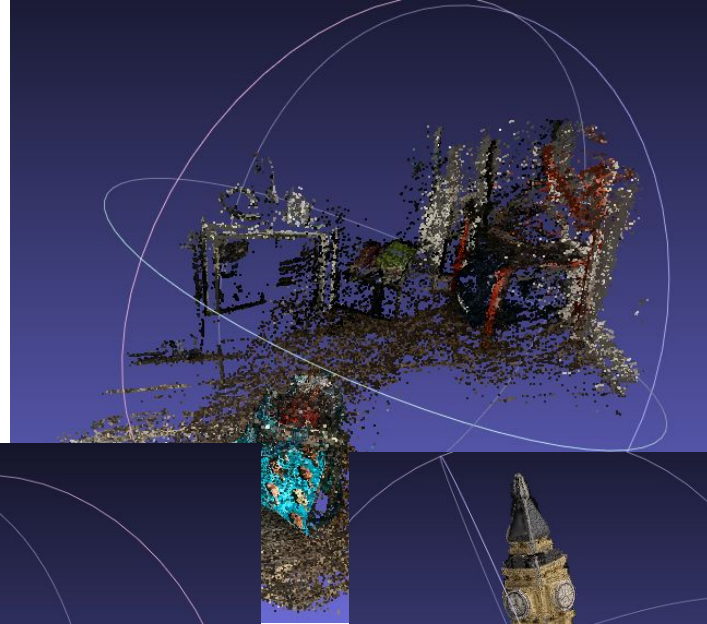
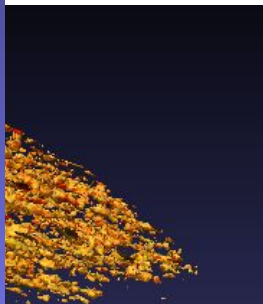
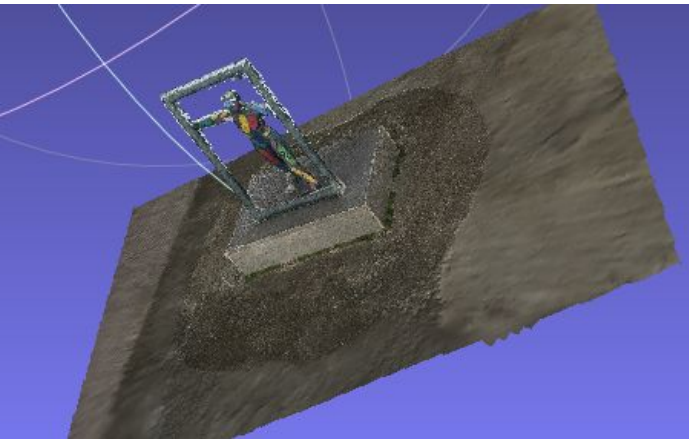


Neuralangelo

Some 3D reconstructions from 2024



Some 3D reconstructions from 2025



Some 3D reconstructions from 2025

